



DAY 7 – Modules & Packages



Goal

Learn how to split Python code into multiple files and reuse it across projects.



Roadmap Notes



What is a Module?

A module is simply a Python file (.py) containing functions, variables, or classes that you can import into another file.

Example

`calculator.py`



```
def add(a, b): return a + b  
def subtract(a, b): return a - b
```

main.py



```
import calculator  
print(calculator.add(5, 3))
```

Output



8

2 Import Specific Functions



```
from calculator import add  
print(add(10, 20))
```

3 Import Everything



```
from calculator import *  
print(add(2,3))
```

⚠ Avoid this in large projects.

4 Import with Alias



```
import calculator as calc  
print(calc.add(5,5))
```

5 Built-in Modules

Python already provides many useful modules.



```
import math import random import os  
import datetime
```

Example



```
import math print(math.sqrt(64))
```

Output



8.0

6 Random Module



```
import random  
print(random.randint(1,100))
```

7 OS Module



```
import os print(os.getcwd())
```

8 Datetime Module



```
import datetime  
print(datetime.datetime.now())
```



What is a Package?

A package is a folder that contains multiple modules.

Example structure



```
project/ main.py utils/ calculator.py  
helper.py
```

Import



```
from utils.calculator import add  
print(add(5,6))
```



Installing Packages



```
pip install numpy
```

Check installed packages



```
pip list
```

Upgrade



```
pip install --upgrade numpy
```



Mini Project

Create



calculator.py main.py

Functions

Add

Subtract

Multiply

Divide

Import all functions into main.py.



Challenge

Create a module named

greetings.py



Inside

```
def hello(name): print(f"Hello {name}")
```



Import it into another file.